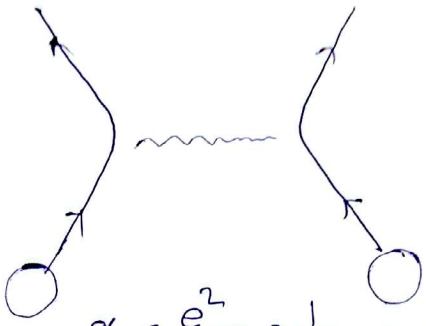
☐ Four fundemental Forces

① Gravitation force

② Electromagnetic force

③ Weak Force

④ Strong Force



$\alpha_e = \frac{e^2}{4\pi\varepsilon_0} = \frac{1}{137} \Rightarrow V(r) = \frac{\alpha_e}{r}$

α_s

α_w

~~Classical Mechanics~~

Particle classifications

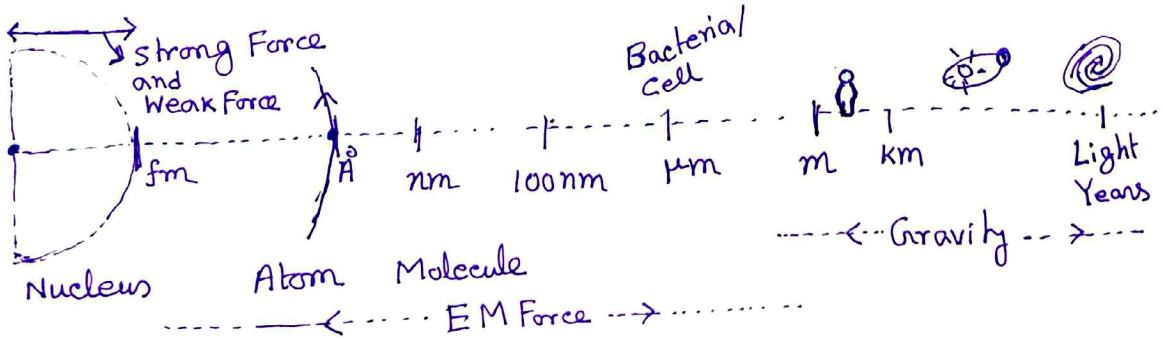
Quarks	Leptons	Gauge Boson
u	e	γ
d	μ	$W^{\pm} Z^0$
s	τ	g
c	ν_e	
b	ν_μ	
t	ν_τ	

H

Mechanics

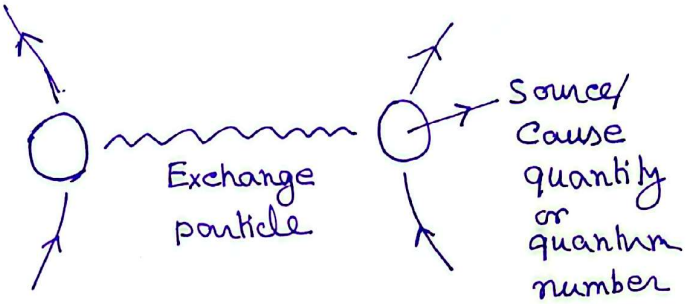
- Classical Mechanics (CM): $F = m\frac{d^2x}{dt^2} = \frac{\alpha_e}{r}$ / .... → $V=0 \Rightarrow E_K = \frac{1}{2}mv^2$; → $V \neq 0, E = E_K + V$
- Relativistic Mechanics (RM): $E_K = \sqrt{p^2c^2 + m_0^2c^4} = mc^2$
- Quantum Mechanics (QM): $E_n = \left(n + \frac{1}{2}\right)h\nu$

These 3 mechanics are language to describe particles, who are facing these four forces. Among the four forces two forces – ① Gravitational force and ② Electromagetic force are mostly working in our daily life. Remaining two forces are associated with nuclear length/time scales.



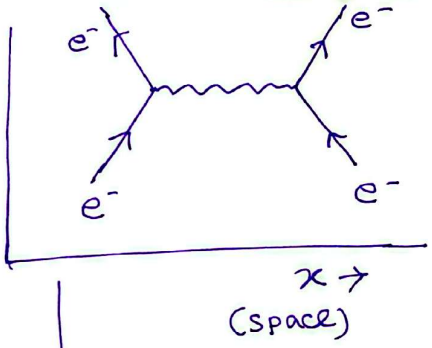
Last 3 forces – ② EM Force, ③ Strong Force and ④ Weak force are redefined in more advance Language (Mathematical framework) – Quantum Field Theory (QFT).

Redefine/New concept of Force → Due to virtual particle exchange



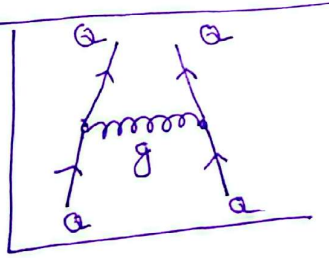
Force	Source/cause	Exchange Particle
② EM Force	electric charge	photon (γ)

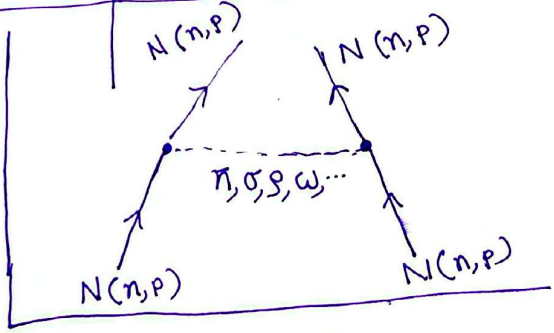
QFT → QED
(Quantum ElectroDynamics)



③ Strong Force

3(A) Nuclear Force	Nuclear charge	Mesons ($\pi, \sigma, \cdots, \rho, \omega, \phi, \cdots$)





3(B) Quark Force	colour charge	Gluons (g)

QFT → QCD
(Quantum chromoDynamics)

④ Weak Force	Weak charge	W^+, W^-, Z^0

2. Electro-magnetic interaction. All charged particles are acted upon by electro-magnetic interaction. Thus electro-magnetic interaction is *charge dependent*. This interaction is attractive as well repulsive.

The *quantum* of electro-magnetic field is a '*photon*'. The inverse square law of Coulomb force ay be conceived of as due to *photon exchange*. The formation of electron-positron pair from a ray photon is an example of electro-magnetic interaction. The binding force between the orbital ectron and the positive nucleus is also due to electro-magnetic interaction.

This force is characterised by a *dimensionless coupling constant*

$$\alpha = \frac{e^2}{4\pi \varepsilon_0 \hbar c} = \frac{1}{137}$$

hich because of its significance in atomic spectroscopy has been termed as *fine structure constant*. has a *characteristic time* of 10^{-20} sec. Its magnitude is much greater than that of gravitational teraction. This interaction is responsible for emission of γ-radiation from the nucleus, all chemical actions, emission of Bremsstrahlung and a number of other phenomenon.

The electro-magnetic interaction also effects uncharged particles that have electric or magnetic oment.

3. Strong interaction. Strong interaction was introduced in order to explain the existence of gable nuclei. This force is of nuclear origin and is strong enough to overcome the repulsive forces etween densely packed protons in the nucleus. This is the dominant interaction which is responsible r most nuclear phenomenon, nuclear energy levels and so on. The strong interaction is independent f electric charge *i.e.* it is the same for $p-p$, $p-n$, and $n-n$ interactions. This strong interaction may e conceived of as due to exchange of pions (π^+, π^0, π^-) or π-mesons.

It is extremely *short range* force (about 10^{-15} *m*) and does not obey inverse square law. It has a *haracteristic time* of 10^{-23} sec. The strength of the nuclear interaction is represented by the magnide of the coupling constant $g^2/\hbar c \approx 15$ where *g* is the pion-nucleon coupling constant.

The strong interaction between *hadrons* should eventually be traceable to an interaction between uarks. The particles that quarks exchange to produce this interaction are called '*gluons*'.

Difference between strong and weak interaction. A characteristic time for nuclear reactions is 0^{-23} sec. All strong interactions take place in about this time. A proton striking a nucleus penetrates n about 10^{-23} sec.

Most of the elementary particles have short lives. For example, hyperons decay into nucleons nd mesons in a time of about 10^{-10} sec. The pions decay into muons in about 10^{-8} sec. These times ppear to be very small but when seen on the nuclear time scale *i.e.* as compared to 10^{-23} sec. these re quite large.

On the other hand radioactive decay of nuclei takes a much longer time. The shortest lived β-ecay requires a time of the order of 10^{-10} sec, while many are known to take seconds, minutes or ours. Had there been strong nuclear or electro-magnetic interaction, there should have been no such elay in the decay process. This delay in the decay process suggests that either these particles are not bject to these forces or there is some new conservative law or prohibition which forbids the decay. ince, eventually the decay takes place, there must be a fourth type of interaction. As the particles ke a long time to respond to such an interaction, the force involved must be very weak as compared strong nuclear force.

This force is called *weak interaction*.

4. Weak interaction. The range of force of weak interaction is less than 10^{-17} *m* and chararistic time 10^{-10} sec. The *field quantum* of weak interaction corresponding to the photon and the on is called '*intermediate vector boson*' of which there seem to be two kinds. Because the weak teraction has such a short range the rest mass of such particles should be large-more than 30 times

the proton mass. One kind called W has a spin of one and charge $\pm e$ and is responsible for ordinary β-decay. The other kind called Z also has a spin one but is electrically neutral and heavier than W. The so called '*intermediate vector boson*' has not been experimentally detected as a free particle.

From Fermi's theory of β-decay we get a numerical constant which is characteristic of weak interaction and the value of which $G_\omega = 1.41 \times 10^{-62}\ \text{Jm}^3$

To get a dimension less constant g_ω we divide G_ω^2 by $[(\hbar c)^2 \times (\text{length})^4]$. A suitable quantity to take for length was found to be rationalised Compton wavelength of the pion *i.e.*

$$\left(\frac{\lambda c}{2\pi}\right)_{\text{pion}} = \frac{\hbar}{m_\pi c} = 1.413 \times 10^{-5}\ \text{m}$$

This gives
$$g_\omega = \left[\frac{G_\omega^2}{(\hbar c)^2}\right]\left[\frac{m_\pi c}{\hbar}\right]^4 \approx 5 \times 10^{-14}$$

The *neutrino* and *anti-neutrino* are exclusively associated with weak interaction just as the *photon* is exclusively associated with electro-magnetic interaction.

Comparison of four fundamental interactions. A comparative chart of four fundamental interactions is given below.

Interaction	*Particles affected*	*Range*	*Relative strength*	*Particle Exchanged*
Strong	Hadrons	$10^{-15}\ m$	1	mesons
Electro-magnetic	Charged particles	∞	$\sim 10^{-2}$	photons
Weak	Hadrons and Leptons	10^{-17} m	$\sim 10^{-13}$	Intermediate vector bosons
Gravitation	All	∞	$\sim 10^{-40}$	Gravitons.

The interaction carrier for all the four interactions are the particles that are exchanged *i.e.* mesons for strong and photons for electro-magnetic interaction

Basic Nuclear properties:

Nuclear magnetic moment:

electron has intrinsic spin s and ~~the~~ electric charge 'e', So magnetic moment μ_s.

For angular momentum $L = m_e v r$, ~~and~~ we can get magnetic moment μ_L due to L.



Now $\frac{\mu_L}{L} = \frac{\frac{e}{T} \times \pi r^2}{m v r}$

$= \frac{e}{2m}$

μ_L = current x area

$= \frac{e}{T} \times \pi r^2$

$v = \frac{2\pi r}{T}$

$\Rightarrow \mu_L = \left(\frac{e\hbar}{2m}\right)\left(\frac{L}{\hbar}\right)$ as $L = \#\hbar$ in microscopic quantum scale.

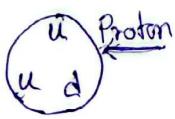
$= \mu_B \left(\frac{L}{\hbar}\right)$ This $\boxed{\mu_B = \frac{e\hbar}{2m}}$ is known as Bohr magneton.

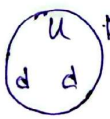
Similarly $\frac{\mu_s}{S} = g_s \left(\frac{e}{2m}\right)$

$\Rightarrow \mu_s = g_s \mu_B \left(\frac{S}{\hbar}\right)$ where $g_s = 2$ is spin g-factor. and $s = \#\hbar$ is spin angular momentum of electron.

Similar to electron, proton can have spin magnetic moment

$\mu_s = 2 \beta_n \left(\frac{S}{\hbar}\right)$. Here $\boxed{\beta_n = \frac{e\hbar}{2m_p}}$ is called nuclear magneton (same as ~~proton's~~ Bohr magneton quantity but for proton)





Spin angular moment of proton is $\mu_p = +2.792 \beta_n$ (Why?)

" " " " neutron " $\mu_n = -1.913 \beta_n$